New Convex Relaxations for Global Optimization Problems in Water Networks[1]

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Outline

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Context and Scope

- Water scarcity is a global issue, mainly caused by overpopulation and climate change.
- The water stress level, which is a demand-driven water scarcity measure (Kummu et al., 2016), is high for many countries.
- The decision making problems in water distribution networks can be categorized into three levels (De Corte and Sörensen, 2016):
  - Strategic level: Network topology.
  - Tactical level: Pipe sizing and pump/valve placement (design).
  - Operational level: Control of pipe and pump/valve.
- The problems in water networks has challenging aspects including
  - Nonconvexity of hydraulics relationships
  - Discrete set of candidate pipes
  - Discrete on-off decisions for pumps and valves

Literature and Our Approach

Existing methods:

- Iterative Global Gradient Algorithm (Todini and Pilati, 1988)
- Local optimization (Burgschweiger et al., 2009).
- Polyhedral outer relaxations (Bonvin et al., 2021).
- Linear/piecewise linear approximations (Verleye and Aghezzaf, 2013; Tasseff et al., 2024).
  - Continuous variable diving heuristic (Choudhary et al., 2026)

Our approach:

We propose a solution method based on **mixed-integer power (second order) cone programming** relaxation built upon the convex hull descriptions of the nonconvex relationships arising from pipes, pumps and valves.

Outline

1 Introduction

2 Model Formulation

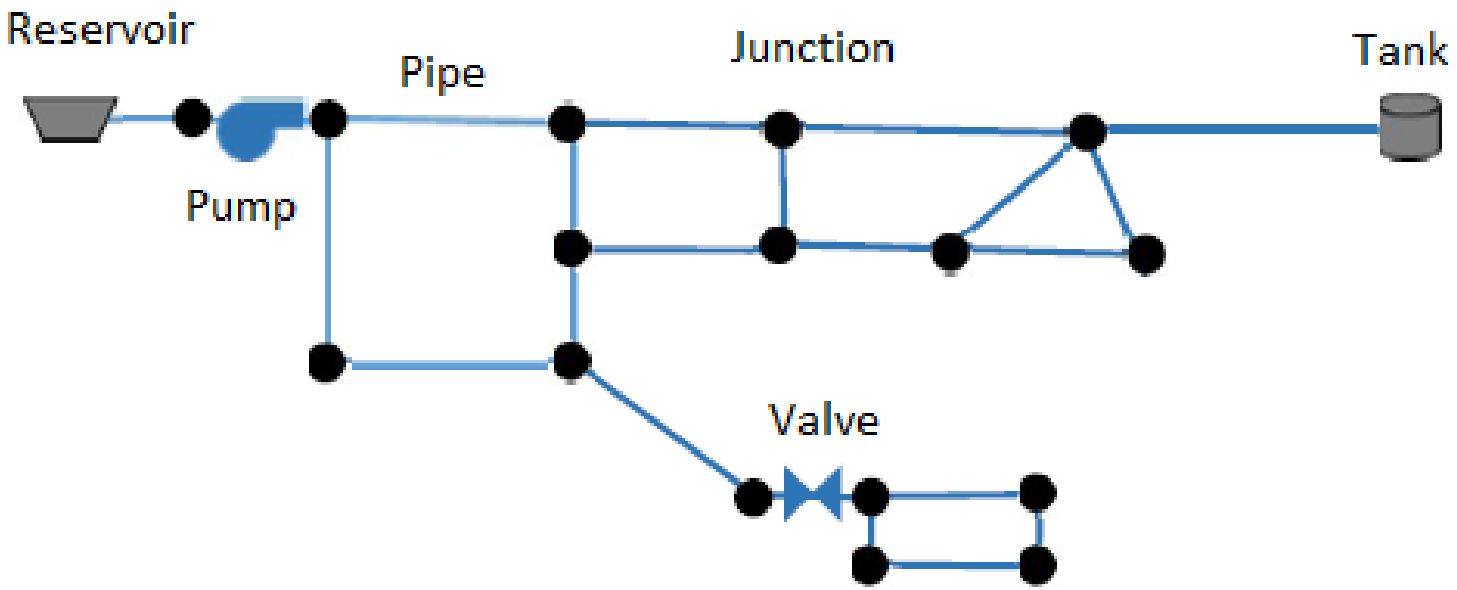
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An Example Water Network



$\mathcal{J}, \mathcal{R}, \mathcal{W}$	Junctions, reservoirs, tanks	δ_i^+	Arcs leaving vertex i
$\mathcal{V}$	All vertices: $\mathcal{J} \cup \mathcal{R} \cup \mathcal{W}$	δ_i^-	Arcs entering vertex i
$\mathcal{L}, \mathcal{P}, \mathcal{C}$	Pipes, pumps, check valves	$\mathcal{T}$	Periods $\{1, \ldots, \lvert\mathcal{T}\rvert\}$
$\mathcal{A}$	All arcs: $\mathcal{L} \cup \mathcal{P} \cup \mathcal{C}$		

Parameters and Decision Variables

$r_{ij} = \beta_{ij} d_{ij}$	Resistance of an installed pipe or valve.
κ	Hydraulic exponent: 1.852 (HW), 2 (DW).
$a_{ij}, b_{ij}, \rho_{ij}$	Pump head-gain parameters; $b_{ij} > 0$, $\rho_{ij} > 1$.
$\underline{f}_{ijt}, \overline{f}_{ijt}$	Active-state flow bounds; nonnegative lower bounds for pumps and check valves.
$\underline{h}_{it}, \overline{h}_{it}$	Head bounds, including storage limits at tanks.
$c_{ijt}, \tilde{c}_{ijt}$	Flow-dependent and fixed pump costs in period t.
g_{it}, Δ	Net junction demand ($g_{it} > 0$ for demand), period duration.
$o_i, h_i^0, \underline{s}_i, \overline{s}_i$	Tank area, empty-tank head, and storage limits.
f_{ijt}	Flow along arc $(i,j) \in \mathcal{A}$ in period $t \in \mathcal{T}$.
h_{it}	Head of vertex $i \in \mathcal{V}$ in period $t \in \mathcal{T}$.
x_{ijt}	1 if arc $(i,j) \in \mathcal{P} \cup \mathcal{C}$ is on in period $t \in \mathcal{T}$, and 0 otherwise.

The Hydraulic Loss in Pipes

The same signed-power relation covers both hydraulic laws:

$$h_i - h_j = r_{ij}\,\mathrm{sign}(f_{ij})|f_{ij}|^{\kappa}, \qquad r_{ij} := \beta_{ij} d_{ij}. \tag{1}$$

$$\text{Hazen–Williams:} \quad r_{ij} = \frac{10.7 L_{ij}}{k_{ij}^{1.852} D_{ij}^{4.87}}, \qquad \kappa = 1.852.$$

L_{ij}: length; D_{ij}: diameter; k_{ij}: roughness coefficient; τ_{ij}: friction factor; g: gravitational acceleration.

- Operational problem: r_{ij} is fixed.
- Design problem: choose r_{ijm} from the finite candidate set $\mathcal{M}_{ij}$.

Pipe, Pump, and Check-Valve Hydraulics

Pipes: always active, signed flow, hydraulic loss

$$h_{it} - h_{jt} = r_{ij}\,\text{sign}(f_{ijt})|f_{ijt}|^{\kappa}, \qquad (i,j) \in \mathcal{L}.$$

Pumps: unidirectional, switchable, head-gain

$$(h_{it} - h_{jt} + a_{ij}x_{ijt} - b_{ij}f_{ijt}^{\rho_{ij}})x_{ijt} = 0, \quad f_{ijt} \geq 0 \qquad (i,j) \in \mathcal{P}.$$

Valves: unidirectional, hydraulic loss when open, zero flow when closed

$$(h_{it} - h_{jt} - r_{ij}f_{ijt}^{\kappa})x_{ijt} = 0, \quad f_{ijt} \geq 0 \qquad (i,j) \in \mathcal{C}.$$

Both switchable components also satisfy

$$\underline{f}_{ijt}x_{ijt} \leq f_{ijt} \leq \overline{f}_{ijt}x_{ijt}, \qquad \underline{f}_{ijt} \geq 0.$$

At $x_{ijt} = 0$, flow is zero and the endpoint heads are decoupled.

Operational Problem MINLP Formulation

$$\min \sum_{(i,j)\in\mathcal{P}} \sum_{t\in\mathcal{T}} \left(c_{ijt} f_{ijt} + \tilde{c}_{ijt} x_{ijt}\right) \tag{2a}$$

subject to

$$h_{it} - h_{jt} = r_{ij}\,\text{sign}(f_{ijt})|f_{ijt}|^{\kappa} \qquad (i,j)\in\mathcal{L}, t\in\mathcal{T} \tag{2b}$$

$$\underline{f}_{ijt} \le f_{ijt} \le \overline{f}_{ijt} \qquad (i,j)\in\mathcal{L}, t\in\mathcal{T} \tag{2c}$$

$$(\underline{h}_{it} - \overline{h}_{jt})(1-x_{ijt}) \le h_{it} - h_{jt} + a_{ij}x_{ijt} - b_{ij} f_{ijt}^{\rho_{ij}} \le (\overline{h}_{it} - \underline{h}_{jt})(1-x_{ijt}) \qquad (i,j)\in\mathcal{P}, t\in\mathcal{T} \tag{2d}$$

$$(\underline{h}_{it} - \overline{h}_{jt})(1-x_{ijt}) \le h_{it} - h_{jt} - r_{ij}\,\text{sign}(f_{ijt})|f_{ijt}|^{\kappa} \le (\overline{h}_{it} - \underline{h}_{jt})(1-x_{ijt}) \qquad (i,j)\in\mathcal{C}, t\in\mathcal{T} \tag{2e}$$

$$\underline{f}_{ijt} x_{ijt} \le f_{ijt} \le \overline{f}_{ijt} x_{ijt} \qquad (i,j)\in\mathcal{P}\cup\mathcal{C}, t\in\mathcal{T} \tag{2f}$$

$$\underline{h}_{it} \le h_{it} \le \overline{h}_{it} \qquad i\in\mathcal{J}\cup\mathcal{R}, t\in\mathcal{T} \tag{2g}$$

$$\sum_{(j,i)\in\delta_i^-} f_{jit} - \sum_{(i,j)\in\delta_i^+} f_{ijt} = g_{it} \qquad i\in\mathcal{J}, t\in\mathcal{T} \tag{2h}$$

$$\Delta\Big(\sum_{(j,i)\in\delta_i^-} f_{jit} - \sum_{(i,j)\in\delta_i^+} f_{ijt}\Big) = o_i(h_{i,t+1} - h_{it}) \qquad i\in\mathcal{W}, t\in\mathcal{T} \tag{2i}$$

$$\underline{h}_{it} \le h_{it} \le \overline{h}_{it} \qquad i\in\mathcal{W}, t\in\mathcal{T}\cup\{|\mathcal{T}|+1\} \tag{2j}$$

$$x_{ijt}\in\{0,1\} \qquad (i,j)\in\mathcal{P}\cup\mathcal{C}, t\in\mathcal{T}. \tag{2k}$$

Design Problem MINLP Formulation

Design Problem is usually modeled over a single representative period ($|\mathcal{T}| = 1$).

$$\min \sum_{(i,j)\in\mathcal{L}} \sum_{m\in\mathcal{M}_{ij}} \hat{c}_{ijm} y_{ijm} \tag{3a}$$

subject to

$$h_{it} - h_{jt} = \left(\sum_{m\in\mathcal{M}_{ij}} r_{ijm} y_{ijm} \right) \text{sign}(f_{ijt}) |f_{ijt}|^{\kappa} \qquad (i,j) \in \mathcal{L}, t \in \mathcal{T} \tag{3b}$$

$$\sum_{m\in\mathcal{M}_{ij}} \underline{f}_{ijmt} y_{ijm} \leq f_{ijt} \leq \sum_{m\in\mathcal{M}_{ij}} \overline{f}_{ijmt} y_{ijm} \qquad (i,j) \in \mathcal{L}, t \in \mathcal{T} \tag{3c}$$

$$\sum_{m\in\mathcal{M}_{ij}} y_{ijm} = 1 \qquad (i,j) \in \mathcal{L} \tag{3d}$$

$$y_{ijm} \in \{0,1\} \qquad (i,j) \in \mathcal{L}, m \in \mathcal{M}_{ij} \tag{3e}$$

(2d)–(2k).

$\mathcal{M}_{ij}$	Set of candidate pipe types for $(i,j) \in \mathcal{L}$.
$r_{ijm} = \beta_{ij} d_{ij}$	Resistance of an installed pipe or valve.
y_{ijm}	1 if candidate type $m \in \mathcal{M}_{ij}$ is selected for pipe $(i,j) \in \mathcal{L}$, and 0 otherwise.

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Definitions

Definition (Power cone)

For $\alpha \in (0,1)$, the three-dimensional power cone (POW) is defined as

$$\mathcal{K}_3^\alpha := \{x \in \mathbb{R}_+^2 \times \mathbb{R} : x_1^\alpha x_2^{1-\alpha} \geq |x_3|\}.$$

Definition (Second-order cone)

The three-dimensional second-order cone (SOC), also known as the Lorentz cone, is defined as

$$\mathcal{O}_3 := \{x \in \mathbb{R}^2 \times \mathbb{R}_+ : x_1^2 + x_2^2 \leq x_3^2\}.$$

Definition (POW and SOC representable sets)

Let $\mathcal{X} \subseteq \mathbb{R}^n$ be a given set. If there exist matrices A and B, and a vector c such that

$$\mathcal{X} = \{x \in \mathbb{R}^n : y \in \mathbb{R}^m : Ax + By + c \in K\},$$

where K is the product of power cones (second order cones), then we call the set $\mathcal{X}$ as power cone (second order cone) representable (**POWr (SOCr)**).

Convex Hull of Sign Power Functions

Let $a \in \mathbb{R}$, $b > 0$, $c > 1$, $\emptyset \neq \mathcal{X} \subseteq \{0,1\}$, and $\underline{y} \leq \overline{y}$ in $\mathbb{R}^3$.

$$\mathcal{H} = [\underline{y}_1, \overline{y}_1] \times [\underline{y}_2, \overline{y}_2] \times [\underline{y}_3, \overline{y}_3],$$
$$\mathcal{H}^0 = [\underline{y}_1, \overline{y}_1] \times [\underline{y}_2, \overline{y}_2] \times [\min\{0, \underline{y}_3\}, \max\{0, \overline{y}_3\}].$$

$$\mathcal{Y}^{a,b,c}(\mathcal{H}, \mathcal{X}) = \left\{ (y, x) \in \mathcal{H}^0 \times \mathcal{X} : \begin{array}{l} (y_1 - y_2 + ax - b\,\mathrm{sign}(y_3)|y_3|^c)x = 0, \\ \underline{y}_3 x \leq y_3 \leq \overline{y}_3 x \end{array} \right\}.$$

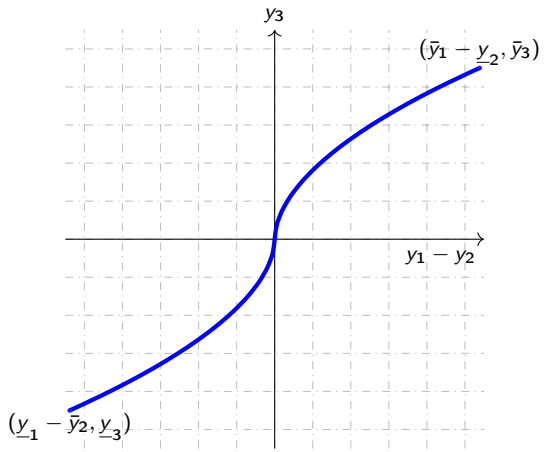
- $y = (h_{it}, h_{jt}, f_{ijt})$ for a network arc and period.
- $\mathcal{H}$ is the active-state box; $\mathcal{H}^0$ also includes zero flow.
- $\mathcal{X} = \{1\}$: always on; $\mathcal{X} = \{0\}$: fixed off; $\mathcal{X} = \{0,1\}$: switchable.

The Three Disjuncts

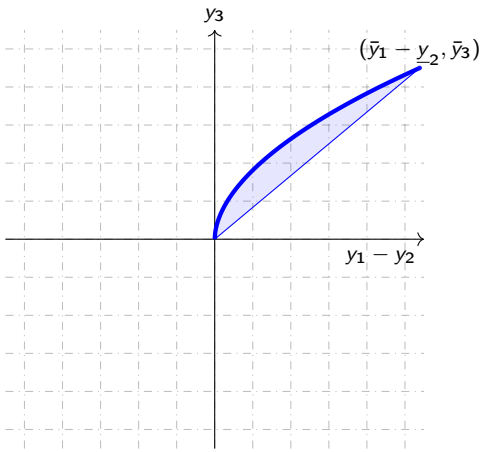
$$\mathcal{Y}_+^{a,b,c}(\mathcal{H}) = \{y \in \mathcal{H} : y_1 - y_2 + a = b y_3^c,\ y_3 \geq 0\},$$
$$\mathcal{Y}_-^{a,b,c}(\mathcal{H}) = \{y \in \mathcal{H} : y_2 - y_1 - a = b(-y_3)^c,\ y_3 \leq 0\},$$
$$\mathcal{Y}_0(\mathcal{H}^0) = \{y \in \mathcal{H}^0 : y_3 = 0\}.$$

$$\mathcal{Y}^{a,b,c}(\mathcal{H}, \mathcal{X}) = (\mathcal{Y}_0(\mathcal{H}^0) \times \{0\}) \cup (\mathcal{Y}_+^{a,b,c}(\mathcal{H}) \times \{1\}) \cup (\mathcal{Y}_-^{a,b,c}(\mathcal{H}) \times \{1\}). \quad (4)$$

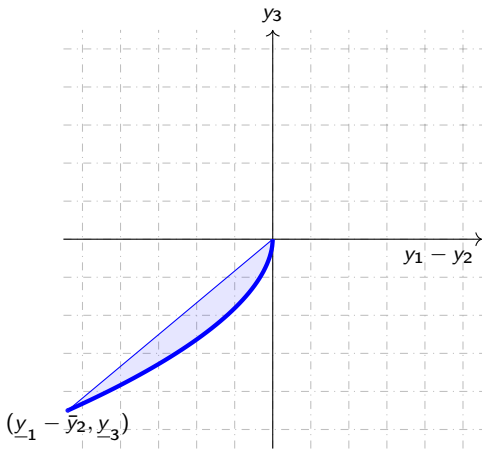
Figure: Some sets in the $(y_1 - y_2, y_3)$ projected space.



(a) Projection of $\mathcal{Y}$ onto the $(y_1 - y_2, y_3)$ space.



(b) Convex hull of the projection of $\mathcal{Y}_+$ onto the $(y_1 - y_2, y_3)$ space.



(c) Convex hull of the projection of $\mathcal{Y}_-$ onto the $(y_1 - y_2, y_3)$ space.

Lemma for Positive Flow: $\text{conv}(\mathcal{Y}_+^{a,b,c}(\mathcal{H}))$

$$\mathcal{Y}_+^{a,b,c}(\mathcal{H}) = \{y \in \mathcal{H} : y_1 - y_2 + a = by_3^c,\ y_3 \geq 0\}$$

Theorem (Theorem 1 in Tawarmalani and Richard (2013))

Consider a continuous function $f : \mathbb{R}^n \to \mathbb{R}$ and a convex set $X \subseteq \mathbb{R}^n$. Define $T = \{x \in \mathbb{R}^n : f(x) = 0, x \in X\}$, $T^{\geq} = \{x \in \mathbb{R}^n : f(x) \geq 0, x \in X\}$, and $T^{\leq} = \{x \in \mathbb{R}^n : f(x) \leq 0, x \in X\}$. Then, $\text{conv}(T) = \text{conv}(T^{\geq}) \cap \text{conv}(T^{\leq})$.

Lemma

Let $a \in \mathbb{R}, b > 0, c > 1$ and $\overline{y}_3 \geq 0$. Then, we have

$$\text{conv}(\mathcal{Y}_+^{a,b,c}(\mathcal{H})) = \{y \in \check{\mathcal{B}}_+^{a,b,c}(\mathcal{H}) : y_1 - y_2 + a \geq by_3^c\},$$

where $\check{\mathcal{B}}_+^{a,b,c}(\mathcal{H})$ is a polytope. Moreover, $\text{conv}(\mathcal{Y}_+^{a,b,c}(\mathcal{H}))$ is POWr.

By Theorem, we have

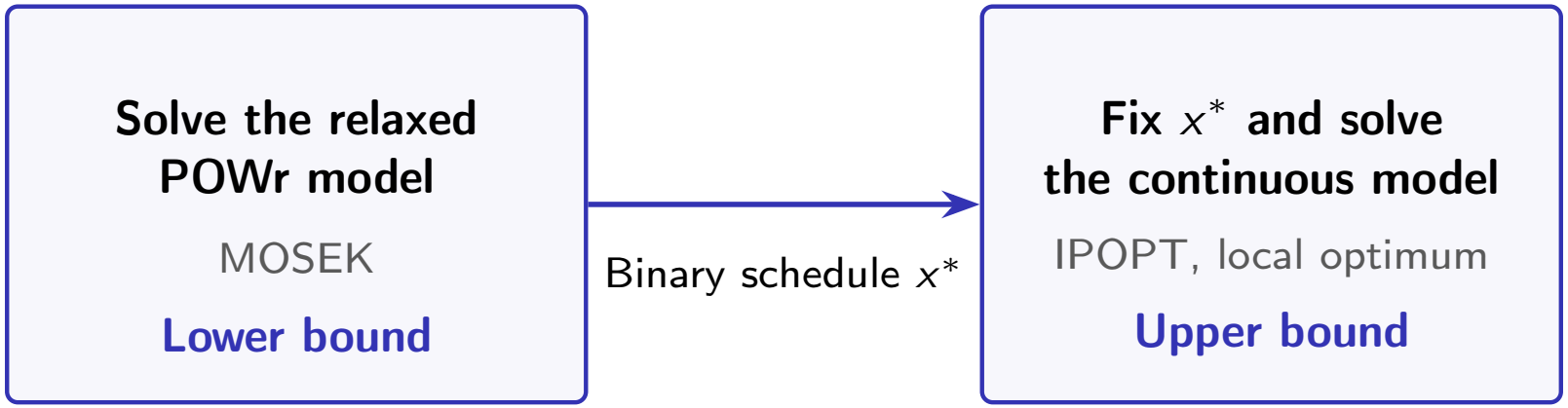
$$\begin{aligned}\text{conv}(\mathcal{Y}_+^{a,b,c}(\mathcal{H})) = \text{conv}(\{y \in \mathcal{H} : y_1 - y_2 + a \geq by_3^c, y_3 \geq 0\}) \\ \cap \text{conv}(\{y \in \mathcal{H} : y_1 - y_2 + a \leq by_3^c, y_3 \geq 0\}).\end{aligned}$$

Algorithm: Extreme points of $\check{\mathcal{B}}_+^{a,b,c}(\mathcal{H})$

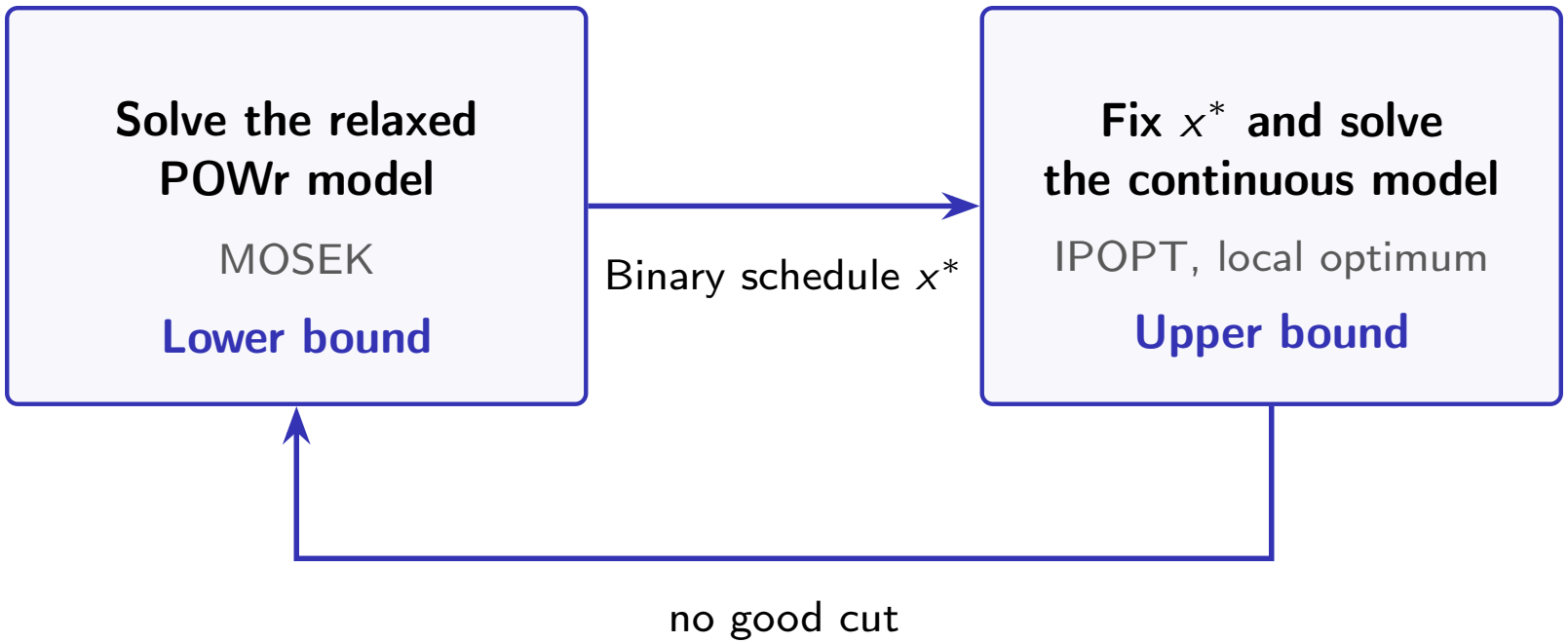
Require: $\underline{y}_1, \overline{y}_1, \underline{y}_2, \overline{y}_2, \underline{y}_3, \overline{y}_3, a, b, c$ with $a \in \mathbb{R}$, $b > 0$, $c > 1$, $\underline{y} \leq \overline{y}$, and $\overline{y}_3 \geq 0$.
Ensure: $\mathcal{Q}_+$

$\mathcal{Q}_+ = \emptyset$, $\ell = \max\{0, \underline{y}_3\}$
for $\tilde{y}_2 \in \{\underline{y}_2, \overline{y}_2\}$ **do**
  for $\tilde{y}_3 \in \{\ell, \overline{y}_3\}$ **do**
    Compute $y' = \min\{\overline{y}_1, \tilde{y}_2 - a + b\tilde{y}_3^c\}$
    if $\underline{y}_1 \leq y'$ **then**
      $\mathcal{Q}_+ = \mathcal{Q}_+ \cup \{(\underline{y}_1, \tilde{y}_2, \tilde{y}_3), (y', \tilde{y}_2, \tilde{y}_3)\}$
for $\tilde{y}_1 \in \{\underline{y}_1, \overline{y}_1\}$ **do**
  for $\tilde{y}_3 \in \{\ell, \overline{y}_3\}$ **do**
    Compute $y' = \max\{\underline{y}_2, \tilde{y}_1 + a - b\tilde{y}_3^c\}$
    if $y' \leq \overline{y}_2$ **then**
      $\mathcal{Q}_+ = \mathcal{Q}_+ \cup \{(\tilde{y}_1, y', \tilde{y}_3), (\tilde{y}_1, \overline{y}_2, \tilde{y}_3)\}$
for $\tilde{y}_1 \in \{\underline{y}_1, \overline{y}_1\}$ **do**
  for $\tilde{y}_2 \in \{\underline{y}_2, \overline{y}_2\}$ **do**
    Compute $y' = \max\left\{\ell, [\max\{0, (\tilde{y}_1 - \tilde{y}_2 + a)/b\}]^{1/c}\right\}$
    if $y' \leq \overline{y}_3$ **then**
      $\mathcal{Q}_+ = \mathcal{Q}_+ \cup \{(\tilde{y}_1, \tilde{y}_2, y'), (\tilde{y}_1, \tilde{y}_2, \overline{y}_3)\}$
return $\mathcal{Q}_+$

Two Phased Solution Method



Iterative Solution Method



No Good Cut

$$\sum_{\substack{(i,j)\in\mathcal{P}\cup\mathcal{C}\\ t\in\mathcal{T},\ x^*_{ijt}=0}} x_{ijt} + \sum_{\substack{(i,j)\in\mathcal{P}\cup\mathcal{C}\\ t\in\mathcal{T},\ x^*_{ijt}=1}} (1 - x_{ijt}) \geq 1.$$

Single-SOC Outer Approximation of Power Cones

Every propoer cone is enclosed by a unique ellipsoidal cone with minimal canonical volume (Seeger, 2016).

Proposition

For $0 < \alpha < \frac{1}{2}$, minimum-canonical-volume SOC containing $\mathcal{K}_3^\alpha$ is $\mathcal{K}_3^{\hat{\alpha}}$.

$$\mathcal{K}_3^\alpha := \{x \in \mathbb{R}^2_+ \times \mathbb{R} : x_1^\alpha x_2^{1-\alpha} \geq |x_3|\}.$$

$$\widehat{\mathcal{K}}_3^\alpha = \left\{ (x_1, x_2, x_3) \in \mathbb{R}^3 : \begin{pmatrix} \sqrt{(1+\alpha)(2-\alpha)}x_1 - (1+\alpha)x_2 \\ \sqrt{3}\left(\frac{(1+\alpha)(2-\alpha)}{\alpha^2}\right)^{\alpha/2} x_3 \\ \sqrt{(1+\alpha)(2-\alpha)}x_1 + (2-\alpha)x_2 \end{pmatrix} \in \mathcal{O}_3 \right\}.$$

$$\max_{\substack{u_1, u_2 \in \mathbb{R}_+ \\ u_1 + u_2 < 1}} \left\{ (1 - u_1 - u_2) \inf_{t \in \mathbb{R}_{++}} \frac{u_1 t^2 + 2\sqrt{(1-u_1)(1-u_2)}t + u_2}{t^{2\alpha}} \right\}.$$

Exact-SOC Representation of POW with Rational Exponent

Exact SOC representation (Ben-Tal and Nemirovski, 2001, Section 3.3.1).
Weighted geometric mean construction (Wang, 2024).

Proposition

Let p, q be coprime positive integers with $q > p$, and let $\hat{k} = \lceil \log_2 q \rceil$. For $x \in \mathbb{R}^2_+ \times \mathbb{R}$, we have $x \in \mathcal{K}_3^{p/q}$ if and only if there exist $\nu_0, \ldots, \nu_{\hat{k}-1} \geq 0$ satisfying $-\nu_0 \leq x_3 \leq \nu_0$ and

$$\left(2\nu_{k-1}, \Pi_{k,1} - \Pi_{k,2}, \Pi_{k,1} + \Pi_{k,2}\right) \in \mathcal{O}_3, \qquad k = 1, \ldots, \hat{k}, \tag{5}$$

where the factors $\Pi_{k,1}, \Pi_{k,2}$ are specified by Algorithm below.

Require: Coprime positive integers p, q with $q > p$.

1: $\hat{k} \leftarrow \lceil \log_2 q \rceil$; introduce $\nu_0, \ldots, \nu_{\hat{k}-1} \geq 0$
2: $(a_1, a_2, a_3) \leftarrow (x_1, x_2, \nu_0)$; $(s_1, s_2, s_3) \leftarrow (p, q-p, 2^{\hat{k}} - q)$
3: **for** $k = \hat{k}, \hat{k}-1, \ldots, 1$ **do**
4: Choose distinct i, j such that s_i, s_j are odd and $s_j \geq s_i$
5: Let $\{l\} = \{1, 2, 3\} \setminus \{i, j\}$
6: $(\Pi_{k,1}, \Pi_{k,2}) \leftarrow (a_i, a_j)$ ▷ SOC at iteration k: $a_i a_j \geq \nu_{k-1}^2$
7: $a_i \leftarrow \nu_{k-1}$; $s_j \leftarrow (s_j - s_i)/2$; $s_l \leftarrow s_l/2$
8: **end for**
9: **return** $\hat{k}$ and $(\Pi_{k,1}, \Pi_{k,2})_{k=1}^{\hat{k}}$

Multiple-SOC Outer Approximation of Power Cones

Proposition

Let $\alpha \in (0,1)$, $\overline{k} \in \mathbb{Z}_{++}$, $b \in \mathbb{R}^3_{++}$, and $u > 0$. Suppose that $\alpha \geq 2^{-\overline{k}}$. Define $\hat{\alpha}$ as the maximum of $\lfloor \alpha q \rfloor / q$ over $q \in \{\lceil 1/\alpha \rceil, \ldots, 2^{\overline{k}}\}$. Then, for every $x \in \mathbb{R}^3$,

$$(b_1 x_1, b_2 x_2, b_3 x_3) \in \mathcal{K}^{\alpha}_3,\ u x_2 \geq |x_3| \implies \left(b_1 x_1, b_2 x_2, \left(\frac{b_3 u}{b_2}\right)^{\frac{\hat{\alpha}}{\alpha}-1} b_3 x_3\right) \in \mathcal{K}^{\hat{\alpha}}_3,\ u x_2 \geq |x_3|. \tag{6}$$

Proof.

By construction, $0 < \hat{\alpha} \leq \alpha$. The assumptions in (6) imply $x_1, x_2 \geq 0$ and

$$(b_1 x_1)^{\alpha}(b_2 x_2)^{1-\alpha} \geq b_3 |x_3|, \qquad \frac{b_3 u}{b_2}(b_2 x_2) \geq b_3 |x_3|. \tag{7}$$

If $\hat{\alpha} = \alpha$, the claim follows immediately. Otherwise, raise the first inequality to $\hat{\alpha}/\alpha$ and the second to $1 - \hat{\alpha}/\alpha$, and multiply them to obtain

$$\left(\frac{b_3 u}{b_2}\right)^{1-\frac{\hat{\alpha}}{\alpha}} (b_1 x_1)^{\hat{\alpha}}(b_2 x_2)^{1-\hat{\alpha}} \geq b_3 |x_3|. \tag{8}$$

Rearranging this inequality and using $x_1, x_2 \geq 0$ proves the $\mathcal{K}^{\hat{\alpha}}_3$ statement in (6). The premise also includes $u x_2 \geq |x_3|$, completing the proof. □

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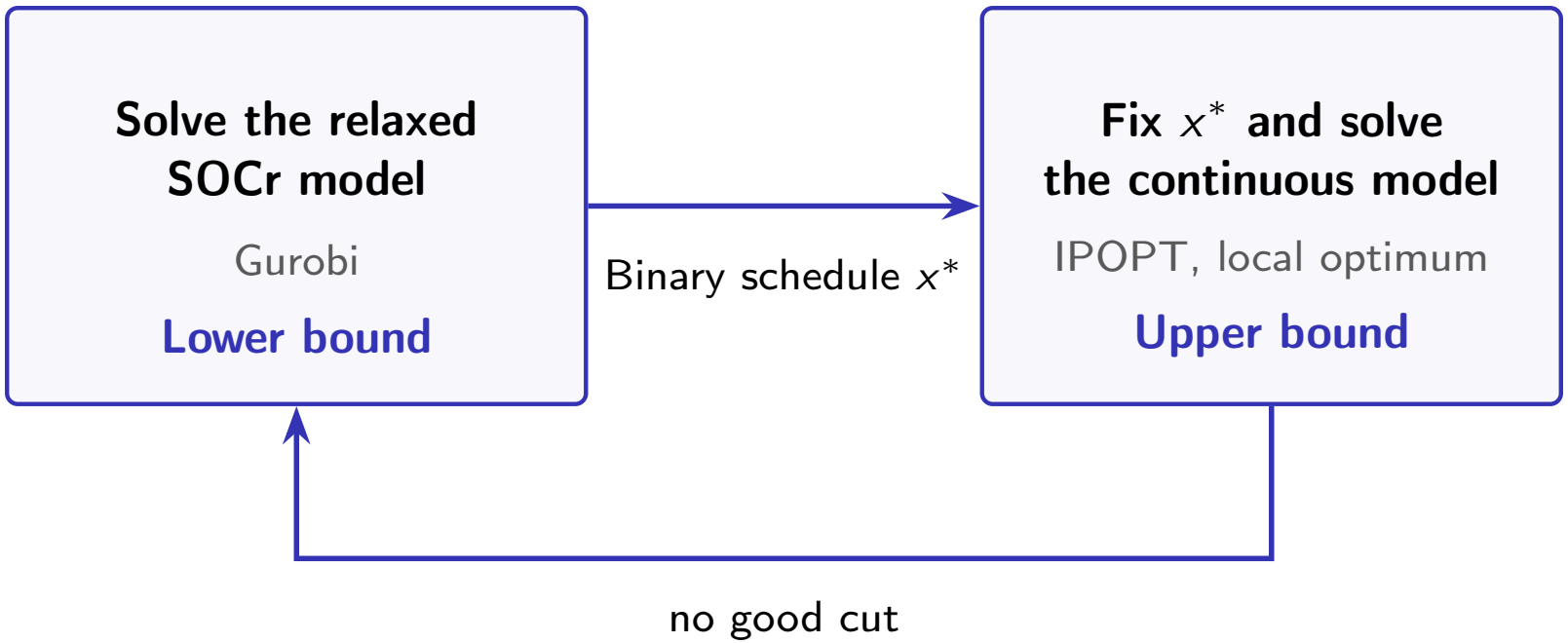
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Callback Framework



No Good Cut

$$\sum_{\substack{(i,j)\in\mathcal{P}\cup\mathcal{C}\\ t\in\mathcal{T},\ x^*_{ijt}=0}} x_{ijt} + \sum_{\substack{(i,j)\in\mathcal{P}\cup\mathcal{C}\\ t\in\mathcal{T},\ x^*_{ijt}=1}} (1 - x_{ijt}) \geq 1.$$

Operational NLP with Fixed Binaries

$$\mathcal{P}_t^{\text{on}} = \left\{(i,j) \in \mathcal{P} : x^*_{ijt} = 1\right\}, \quad \mathcal{C}_t^{\text{on}} = \left\{(i,j) \in \mathcal{C} : x^*_{ijt} = 1\right\}$$

$$\min \sum_{t\in\mathcal{T}} \sum_{(i,j)\in\mathcal{P}_t^{\text{on}}} \left(c_{ijt} f_{ijt} + \tilde{c}_{ijt}\right) \tag{9a}$$

subject to

$$h_{it} - h_{jt} = r_{ij}\,\text{sign}(f_{ijt})|f_{ijt}|^{\kappa} \qquad (i,j) \in \mathcal{L} \cup \mathcal{C}_t^{\text{on}}, t \in \mathcal{T} \tag{9b}$$

$$h_{it} - h_{jt} = b_{ij} f_{ijt}^{\rho_{ij}} - a_{ij} \qquad (i,j) \in \mathcal{P}_t^{\text{on}}, t \in \mathcal{T} \tag{9c}$$

$$\sum_{(j,i)\in\delta_i^-} f_{jit} - \sum_{(i,j)\in\delta_i^+} f_{ijt} = g_{it} \qquad i \in \mathcal{J}, t \in \mathcal{T} \tag{9d}$$

$$\Delta\Big(\sum_{(j,i)\in\delta_i^-} f_{jit} - \sum_{(i,j)\in\delta_i^+} f_{ijt}\Big) = o_i(h_{i,t+1} - h_{i,t}) \qquad i \in \mathcal{W}, t \in \mathcal{T} \tag{9e}$$

$$\underline{h}_{it} \le h_{it} \le \overline{h}_{it} \qquad (i,t) \in (\mathcal{V}\times\mathcal{T}) \cup (\mathcal{W} \times \{|\mathcal{T}|+1\}) \tag{9f}$$

$$\underline{f}_{ijt} \le f_{ijt} \le \overline{f}_{ijt} \qquad (i,j) \in \mathcal{L} \cup \mathcal{P}_t^{\text{on}} \cup \mathcal{C}_t^{\text{on}}, t \in \mathcal{T} \tag{9g}$$

Operational NLP with Fixed Binaries

$$\min \sum_{t\in\mathcal{T}} \sum_{(i,j)\in\mathcal{P}_t^{\mathrm{on}}} \left(c_{ijt} f_{ijt} + \tilde{c}_{ijt}\right) \tag{10a}$$

subject to

$$h_{it} - h_{jt} = r_{ij}\,\mathrm{sign}(f_{ijt})|f_{ijt}|^{\kappa} \qquad (i,j)\in\mathcal{L}\cup\mathcal{C}_t^{\mathrm{on}}, t\in\mathcal{T} \tag{10b}$$

$$h_{it} - h_{jt} = b_{ij} f_{ijt}^{\rho_{ij}} - a_{ij} \qquad (i,j)\in\mathcal{P}_t^{\mathrm{on}}, t\in\mathcal{T} \tag{10c}$$

$$\sum_{(j,i)\in\delta_i^-} f_{jit} - \sum_{(i,j)\in\delta_i^+} f_{ijt} = g_{it} \qquad i\in\mathcal{J}, t\in\mathcal{T} \tag{10d}$$

$$\Delta\Big(\sum_{(j,i)\in\delta_i^-} f_{jit} - \sum_{(i,j)\in\delta_i^+} f_{ijt}\Big) = o_i(h_{i,t+1} - h_{i,t}) \qquad i\in\mathcal{W}, t\in\mathcal{T} \tag{10e}$$

$$\underline{h}_{it} \le h_{it} \le \overline{h}_{it} \qquad (i,t)\in(\mathcal{V}\times\mathcal{T})\cup(\mathcal{W}\times\{|\mathcal{T}|+1\}) \tag{10f}$$

$$\underline{f}_{ijt} \le f_{ijt} \le \overline{f}_{ijt} \qquad (i,j)\in\mathcal{L}\cup\mathcal{P}_t^{\mathrm{on}}\cup\mathcal{C}_t^{\mathrm{on}}, t\in\mathcal{T} \tag{10g}$$

Proposition

Suppose that, for every $t\in\mathcal{T}$, each connected component of $N_t = (\mathcal{V}, \mathcal{L}\cup\mathcal{P}_t^{\mathrm{on}}\cup\mathcal{C}_t^{\mathrm{on}})$ contains a vertex in $\mathcal{R}\cup\mathcal{W}$. Then (10) is either infeasible or has a unique feasible solution.

KKT Conditions of a Convex Program

Define the arc potentials

$$\Phi_{ij}^{L}(f) = \frac{r_{ij}}{\kappa + 1}|f|^{\kappa+1}, \qquad \Phi_{ij}^{P}(f) = \frac{b_{ij}}{\rho_{ij} + 1}|f|^{\rho_{ij}+1} - a_{ij}f.$$

For a fixed period and known reservoir/tank heads, we solve

$$\begin{aligned}
\min_{f} \quad & \sum_{(i,j)\in\mathcal{L}\cup\mathcal{C}_t^{\text{on}}} \Phi_{ij}^{L}(f_{ijt}) + \sum_{(i,j)\in\mathcal{P}_t^{\text{on}}} \Phi_{ij}^{P}(f_{ijt}) \\
& + \sum_{i\in\mathcal{R}\cup\mathcal{W}} h_{it} \left(\sum_{(j,i)\in\delta_i^-} f_{jit} - \sum_{(i,j)\in\delta_i^+} f_{ijt} \right) \\
\text{s.t.} \quad & \sum_{(j,i)\in\delta_i^-} f_{jit} - \sum_{(i,j)\in\delta_i^+} f_{ijt} = g_{it}, \qquad i \in \mathcal{J}.
\end{aligned}$$

- We only define pumps and valves with active status.
- Physical flow/head bounds are checked *after* optimization so their multipliers do not change hydraulic stationarity.

Convex Continuous NLP with Fixed Binaries

Require: Active sets $\mathcal{P}_t^{\text{on}}$ and $\mathcal{C}_t^{\text{on}}$ for every $t \in \mathcal{T}$; prescribed reservoir heads and initial tank heads; every connected component of N_t contains a vertex in $\mathcal{R} \cup \mathcal{W}$

Ensure: The unique feasible flows and heads (f, h), or a declaration that x^* is infeasible

1: **for** $t = 1, \ldots, |\mathcal{T}|$ **do**
2: Solve convex program using the known reservoir and current tank heads; obtain f_{ijt} and λ_{it}, $i \in \mathcal{J}$
3: Set $h_{it} \leftarrow \lambda_{it}$ for every $i \in \mathcal{J}$
4: Set $h_{i,t+\mathbf{1}} \leftarrow h_{it} + \frac{\Delta}{o_i} \left(\sum_{(j,i) \in \delta_i^-} f_{jit} - \sum_{(i,j) \in \delta_i^+} f_{ijt} \right)$ for every $i \in \mathcal{W}$
5: **if** an active flow or a period-t head violates its bound, or an updated tank head $h_{i,t+\mathbf{1}}$ violates its bound **then**
6: Report x^* infeasible and stop
7: **end if**
8: **end for**
9: Return (f, h)

Valid Inequalities for Vertex Subsets

For $S \subseteq \mathcal{J} \cup \mathcal{W}$ and $1 \leq \underline{t} \leq \overline{t} \leq |\mathcal{T}|$,

$$\sum_{t=\underline{t}}^{\overline{t}} \Bigg[\sum_{(i,j)\in\delta^-(\mathcal{S})\cap(\mathcal{P}\cup\mathcal{C})} \overline{f}_{ijt} x_{ijt} - \sum_{(i,j)\in\delta^+(\mathcal{S})\cap(\mathcal{P}\cup\mathcal{C})} \underline{f}_{ijt} x_{ijt} + \sum_{(i,j)\in\delta^-(\mathcal{S})\cap\mathcal{L}} \overline{f}_{ijt} \lambda^+_{ijt} - \sum_{(i,j)\in\delta^+(\mathcal{S})\cap\mathcal{L}} \underline{f}_{ijt}(1-\lambda^+_{ijt}) \Bigg] \geq \sum_{t=\underline{t}}^{\overline{t}} \sum_{i\in S\cap\mathcal{J}} g_{it} + \sum_{i\in S\cap\mathcal{W}} \frac{o_i}{\Delta}(\underline{h}_{i,\overline{t}+1} - \overline{h}_{i,\underline{t}}),$$

$$\sum_{t=\underline{t}}^{\overline{t}} \Bigg[\sum_{(i,j)\in\delta^-(\mathcal{S})\cap(\mathcal{P}\cup\mathcal{C})} \underline{f}_{ijt} x_{ijt} - \sum_{(i,j)\in\delta^+(\mathcal{S})\cap(\mathcal{P}\cup\mathcal{C})} \overline{f}_{ijt} x_{ijt} + \sum_{(i,j)\in\delta^-(\mathcal{S})\cap\mathcal{L}} \underline{f}_{ijt}(1-\lambda^+_{ijt}) - \sum_{(i,j)\in\delta^+(\mathcal{S})\cap\mathcal{L}} \overline{f}_{ijt} \lambda^+_{ijt} \Bigg] \leq \sum_{t=\underline{t}}^{\overline{t}} \sum_{i\in S\cap\mathcal{J}} g_{it} + \sum_{i\in S\cap\mathcal{W}} \frac{o_i}{\Delta}(\overline{h}_{i,\overline{t}+1} - \underline{h}_{i,\underline{t}}).$$

Fixing Flows at Bridges

$$\sum_{(i,j)\in\delta^-(\mathcal{S})} f_{ijt} - \sum_{(i,j)\in\delta^+(\mathcal{S})} f_{ijt} = \sum_{i\in\mathcal{S}\cap\mathcal{J}} g_{it} + \sum_{i\in\mathcal{S}\cap\mathcal{W}} \frac{o_i}{\Delta}(h_{i,t+1} - h_{i,t}) \quad t \in \mathcal{T} \quad (11)$$

Removing the undirected bridge represented by arc (i,j) separates the network into two components: S containing i and S' containing j.

A side containing only junctions fixes the bridge flow

$$S \subseteq \mathcal{J} \implies f_{ijt} = -\sum_{k\in S} g_{kt}, \qquad S' \subseteq \mathcal{J} \implies f_{ijt} = \sum_{k\in S'} g_{kt}.$$

If a side contains tanks but no reservoirs, retain the storage-change term:

$$S \subseteq \mathcal{J}\cup\mathcal{W} \implies f_{ijt} = -\sum_{k\in S\cap\mathcal{J}} g_{kt} - \sum_{k\in S\cap\mathcal{W}} \frac{o_k}{\Delta}(h_{k,t+1} - h_{kt}),$$

$$S' \subseteq \mathcal{J}\cup\mathcal{W} \implies f_{ijt} = \sum_{k\in S'\cap\mathcal{J}} g_{kt} + \sum_{k\in S'\cap\mathcal{W}} \frac{o_k}{\Delta}(h_{k,t+1} - h_{kt}).$$

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- Setting
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Hardware and Sofware

- **Hardware:** 64-bit Apple Silicon MacBook Pro running macOS Tahoe 26.2, equipped with an Apple M3 Pro processor (6 performance and 6 efficiency cores) and 36 GB of unified memory.
- **Language:** Python 3.13.6
- **Optimization solvers:** Gurobi Optimizer 13.0.2, BARON 26.4.12, MOSEK 11.2.0, IPOPT 3.14.19

Preliminary Experiments

ATM-12 benchmark (Tasseff et al., 2022): 22 junctions, 3 reservoirs, 2 tanks, 41 pipes, 3 pumps, and 3 time periods. Five instances (i.e., day) differ in their variable cost of running pumps.

	Gurobi				BARON				Two Phased Algorithm			
Ins.	Time(s)	LB	UB	Gap(%)	Time(s)	LB	UB	Gap(%)	Time(s)	LB	UB	Gap(%)
Day1	82.40	220.67	295.70	25.37*	TL†	199.70	220.66	9.50	1.42	209.67	220.66	4.98
Day2	11.24	184.66	184.68	0.01	TL†	167.33	184.68	9.39	1.16	175.65	184.68	4.88
Day3	73.09	195.91	263.60	25.68*	TL†	178.06	195.91	9.11	1.12	185.56	195.91	5.28
Day4	103.50	216.75	291.32	25.60*	TL†	199.33	216.75	8.04	0.90	204.64	216.75	5.59
Day5	64.82	241.18	317.79	24.11*	TL†	220.96	241.18	8.38	1.12	220.96	241.18	8.38

* Sub-optimal termination (unable to solve some node relaxations).
† The time limit of 600 seconds reached.

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Conclusion and Future Work

- We present a global optimization framework for the Operational Problem in water networks considering
  - nonconvexity of hydraulics equations for pipes
  - nonconvexity of hydraulics equations and discrete on-off decisions for pumps
  - nonconvexity of head-gain and discrete on-off decisions for pumps
  - multi-period nature.
- We obtain the convex hull of the hydraulic equation for pipes and head-gain equation for pumps as power cone representable sets.
- We propose a solution approach based on these convexifications.

Thank you for listening!

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References I

Aharon Ben-Tal and Arkadi Nemirovski. *Lectures on modern convex optimization: analysis, algorithms, and engineering applications*. SIAM, 2001.

Gratien Bonvin, Sophie Demassey, and Andrea Lodi. Pump scheduling in drinking water distribution networks with an lp/nlp-based branch and bound. *Optimization and Engineering*, 22(3):1275–1313, 2021.

Jens Burgschweiger, Bernd Gnädig, and Marc C Steinbach. Optimization models for operative planning in drinking water networks. *Optimization and Engineering*, 10:43–73, 2009.

Sourabh K. Choudhary, Santanu S. Dey, and Nikolaos V. Sahinidis. Water network design and operation optimization: Leveraging approximations. *EURO Journal on Computational Optimization*, 14:100145, 2026. ISSN 2192-4406.

Annelies De Corte and Kenneth Sörensen. An iterated local search algorithm for water distribution network design optimization. *Networks*, 67(3):187–198, 2016.

Matti Kummu, Joseph HA Guillaume, Hans de Moel, Stephanie Eisner, Martina Flörke, Miina Porkka, Stefan Siebert, Ted IE Veldkamp, and PJ Ward. The world's road to water scarcity: shortage and stress in the 20th century and pathways towards sustainability. *Scientific reports*, 6(1):38495, 2016.

Alberto Seeger. Conic volume optimization: canonical measures, ellipsoidal cones, and power cones. *Set-Valued and Variational Analysis*, 24(2):237–262, 2016.

Byron Tasseff, Russell Bent, Carleton Coffrin, Clayton Barrows, Devon Sigler, Jonathan Stickel, Ahmed S Zamzam, Yang Liu, and Pascal Van Hentenryck. Polyhedral relaxations for optimal pump scheduling of potable water distribution networks. *INFORMS Journal on Computing*, 36(4):1040–1063, 2024.

References II

Mohit Tawarmalani and Jean-Philippe P Richard. Decomposition techniques in convexification of inequalities. Technical report, Technical report, 2013.

Ezio Todini and Stefano Pilati. A gradient algorithm for the analysis of pipe networks. In *Computer applications in water supply: vol. 1—systems analysis and simulation*, pages 1–20. 1988.

Derek Verleye and El-Houssaine Aghezzaf. Optimising production and distribution operations in large water supply networks: A piecewise linear optimisation approach. *International Journal of Production Research*, 51(23-24):7170–7189, 2013.

Jie Wang. Weighted geometric mean, minimum mediated set, and optimal simple second-order cone representation. *SIAM Journal on Optimization*, 34(2):1490–1514, 2024.